HW4 Report

1. Part 2

1.1 The Lifecycle of a Cloud Task

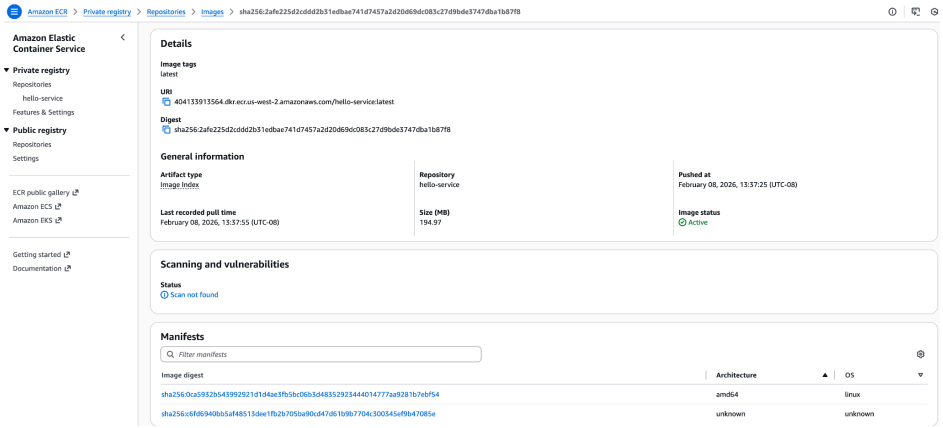
In this phase, I established the "Build-Push-Launch" workflow required for modern cloud-native applications:

1. **Docker image:** build a new docker image based on hw2.
2. **ECR (Elastic Container Registry):** Managed image versions and permissions.
3. **ECS (Elastic Container Service):** Deployed tasks using the serverless Fargate launch type.

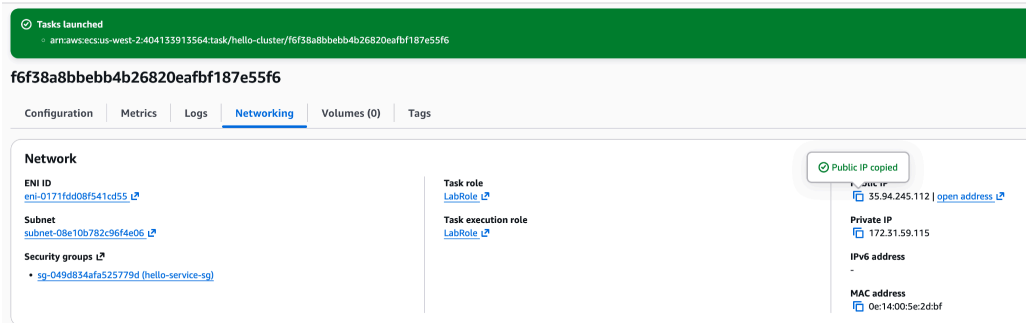
Here are the 7 steps

1. Configure AWS.
2. Go to Amazon ECR to create a repository, then copy the repository URI.
3. Push Docker image to ECR.

```
molly@mollys-MacBook-Pro hw4 % molly@mollys-MacBook-Pro hw4 % aws ecr
 list-images \
  --repository-name hello-service \
  --region us-west-2 \
  --query 'imageIds[*].imageTag' \
  --output table
------------
|ListImages|
+----------+
|  latest  |
+----------+
```



4. Create an ECS Cluster.
5. Create task definition.
6. Run a task.

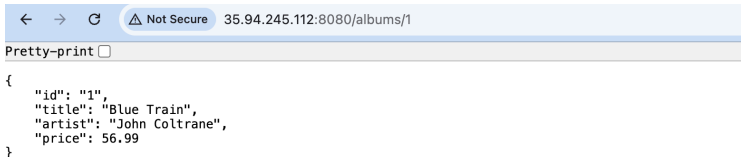


7. Getting the Public IP and Testing.

```
molly@mollys-MacBook-Pro hw4 % curl http://35.94.245.112:8080/albums

[
    {
        "id": "1",
        "title": "Blue Train",
        "artist": "John Coltrane",
        "price": 56.99
    },
    {
        "id": "2",
        "title": "Jeru",
        "artist": "Gerry Mulligan",
        "price": 17.99
    },
    {
        "id": "3",
        "title": "Sarah Vaughan and Clifford Brown",
        "artist": "Sarah Vaughan",
        "price": 39.99
    }
]%
molly@mollys-MacBook-Pro hw4 % curl http://35.94.245.112:8080/albums/
1

{
    "id": "1",
    "title": "Blue Train",
    "artist": "John Coltrane",
    "price": 56.99
}%
```

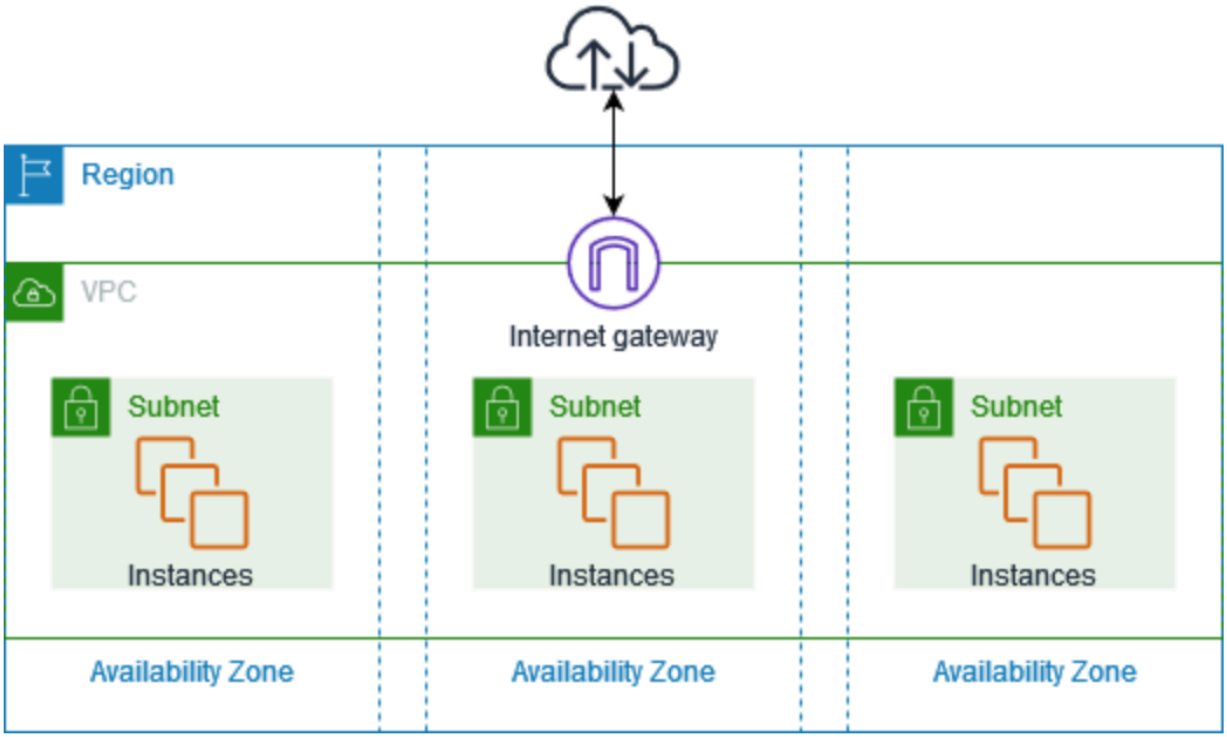


1.2 Technical Concept Exploration

Difference between EC2 and ECS:

- **EC2 (Elastic Compute Cloud):** Provides raw Virtual Machines. You are responsible for managing the OS, patching, and installing Docker yourself. It offers the most control but the highest management overhead.
- **ECS (Elastic Container Service):** An orchestration service. With **Fargate**, it is serverless, meaning AWS manages the underlying servers. You only provide the container image and resource requirements.

VPC and Subnets:



- **VPC (Virtual Private Cloud):** A logically isolated virtual network in AWS. It defines the networking environment for your resources.
- **Subnet:** A range of IP addresses within a VPC. It allows you to group resources based on security and operational needs.
- **Accessing the Default VPC:** AWS provides a default VPC in every region with public subnets. I accessed it by assigning a **Public IP** to my Fargate tasks and configuring **Security Groups** (virtual firewalls) to allow inbound traffic on port 8080.

TCP vs. UDP:

Feature	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Connection	Connection-oriented (Handshake required)	Connectionless (Fire and forget)
Reliability	Guaranteed delivery (Retries on loss)	No guarantee (Packets can be lost)

Ordering	Guarantees packets arrive in sequence	No guarantee of order
Speed	Slower (due to overhead and error checking)	Faster (minimal overhead)
Data Integrity	High (Checksums and acknowledgments)	Low (No error recovery)
Best Used For	Web browsing (HTTP), Email, File transfers	Video streaming, VoIP, Online gaming

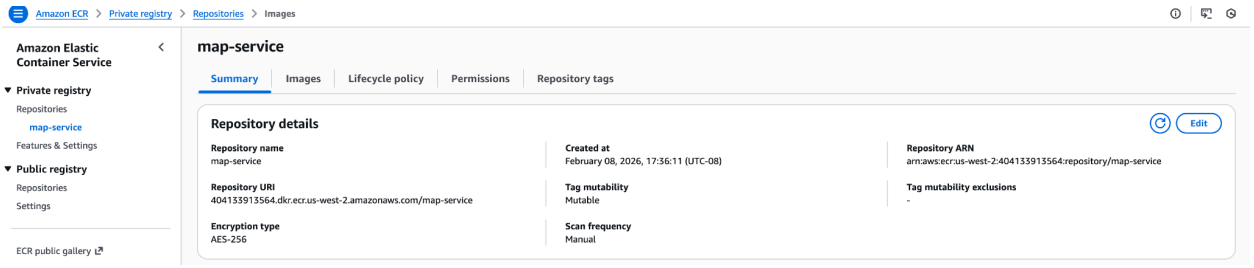
- **TCP (Transmission Control Protocol):** Connection-oriented and reliable. It ensures all packets arrive in the correct order. This is essential for our MapReduce project because a single missing word count would invalidate the results.
- **UDP (User Datagram Protocol):** Connectionless and faster, but does not guarantee delivery. It is used for streaming or gaming where speed is prioritized over 100% data integrity.

Controlling Resource Allocation: Resources are controlled in the **Task Definition**:

- **Task Level:** Sets the total CPU and Memory for all containers in the task.
- **Container Level:** Defines **Hard Limits** (task is killed if exceeded) and **Soft Limits** (guaranteed reservation). This ensures one container doesn't consume all resources on a shared host.

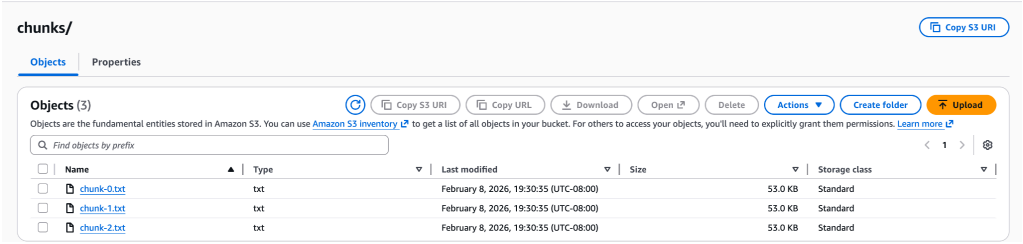
2. Part 3: Distributed MapReduce Implementation

2.1 Implementation Logic

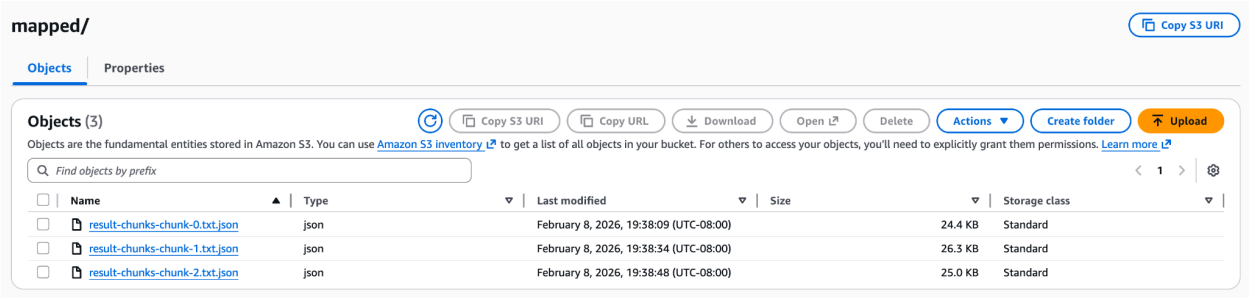


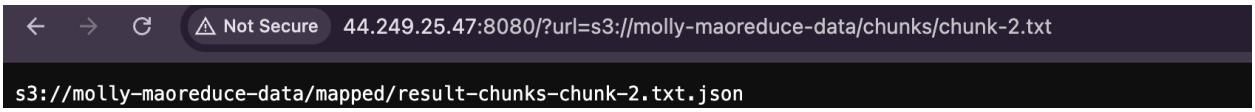
The project followed a **"Single Binary, Multiple Roles"** pattern. I implemented three distinct functions within one Go program, triggered by the APP_ROLE environment variable:

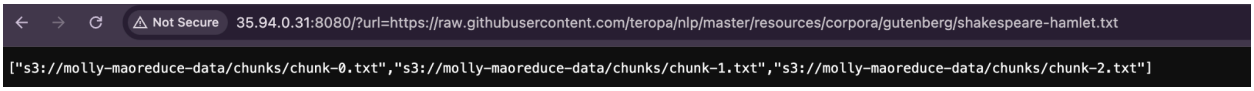
1. **Splitter:** Reads the source text from a URL, calculates split points based on character count while respecting newline boundaries, and uploads three .txt chunks to S3.



2. **Mapper:** Triggered via a URL parameter, it downloads a specific chunk from S3, tokenizes the text, standardizes it (lowercase/no punctuation), and uploads a JSON map of word counts.

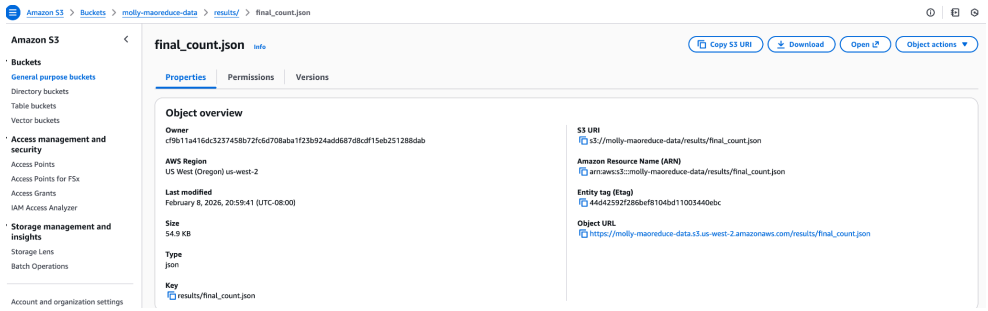






3. **Reducer:** Receives multiple Mapper URLs, downloads all JSON results, aggregates the

counts into a single master map, and saves the final final_count.json.



2.2 Challenges and Debugging

During implementation, I encountered and resolved several critical issues:

- **ECR Tag Immutability:** Overwriting the :latest tag was initially blocked. I had to change the repository settings to **Mutable** to allow for iterative code updates.
- **Silent Failures (Empty Results):** Early runs produced an empty {} result. This was due to the Reducer attempting to process S3 links before the Mappers had finished writing or due to incorrect URL parameter parsing. I added **Robust Logging** and **Error Checks** to ensure the Reducer only proceeds if data is successfully downloaded.
- **Manual Orchestration:** manually passing URLs between tasks felt like a **'Human Relay Race.'** While it helped me understand the data flow, it highlighted the massive **Network and Operational Overhead** inherent in distributed systems. This experience proved to me that for big data, **Automation and Orchestration** (like Step Functions) are not just optional tools—they are absolute necessities.

I noticed that my Reducer is **idempotent** regarding the request execution. If I refresh the browser multiple times, it simply re-calculates and overwrites the same `final_count.json` in S3. While this ensures the final result is correct, it highlights a **resource inefficiency**. In a production system, I would implement a **locking mechanism** (like Redis locks) or check for the existence of the output file to skip redundant processing.

2.3 Verification

I verified the final output using a Python verification script. The counts (e.g., hamlet: 100) matched the source text exactly, proving the distributed logic is sound and the data remained consistent across the S3 hand-offs.

```
(.venv) molly@mollys-MacBook-Pro part3 % python result-verification.py
Downloading file from source for ground truth verification...
URL: https://raw.githubusercontent.com/teropa/nlp/master/resources/corpora/gutenberg/shakespeare-hamlet.txt
Local verification complete! Total word count: 30266

Word         | Ground Truth    | Your System     | Status
=================================================================
hamlet       | 100             | 100             | ✅ Match
hand         | 19              | 19              | ✅ Match
lord         | 211             | 211             | ✅ Match
queen        | 8               | 8               | ✅ Match
horatio      | 40              | 40              | ✅ Match

Conclusion: Your system's results match the source file perfectly! The counts are small because the source file itself is small.
```

2.4 Performance Analysis

- Comparative Data

Metric	Sequential (Single Task)	Distributed (3 Mappers)
Total Latency	~8.3 seconds	~12.8 seconds
Data Transferred	Fetch Once (GitHub)	Fetch + 6x S3 I/O Ops
Compute Cost	0.25 vCPU	3x 0.25 vCPU (Burst)

- Analysis

  In a production environment, MapReduce is expected to be faster. However, for a file of this size, we observe **Negative Scaling**. The reasons are:

  1. **Orchestration Latency:** The "Human-in-the-loop" factor. Manually copying URLs between browser tabs introduces several seconds of delay that wouldn't exist in an automated system.
  2. **Network & I/O Overhead:** In the sequential version, data is read once. In the distributed version, data is uploaded/downloaded across S3 7 times (1 Splitter upload, 3 Mapper downloads, 3 Mapper uploads). For small files, the time spent on S3 API calls is greater than the compute time.